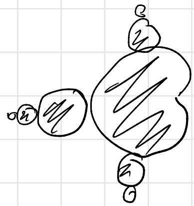
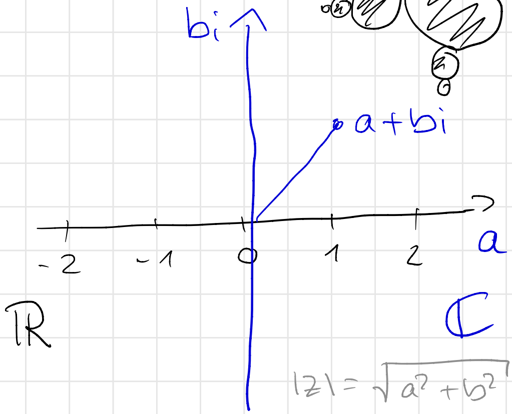
Mandelbrot





→ a set of numbers

$f_c(z) = z^2 + c$

c is a **Complex Number**

$a + bi$

$i^2 = -1$

$x^2 + 1 = 0$

$i = \sqrt{-1}$

$$z \rightarrow z^2 + c$$

- Chose a c
- z starts at 0, then we iterate

$c = 1 + 0i = 1$

$f_1(0) = 0^2 + 1 = 1$

$f_1(1) = 1^2 + 1 = 2$

$f_1(2) = 2^2 + 1 = 5$

$f_1(5) = 5^2 + 1 = 26$

$f_1(26) = 26^2 + 1 = 676$

CASE 1: $|z|$ **unbounded**
(escapes to infinity)

$c = -1 + 0i = -1$

$f_{-1}(0) = 0^2 - 1 = -1$

$f_{-1}(-1) = (-1)^2 - 1 = 0$

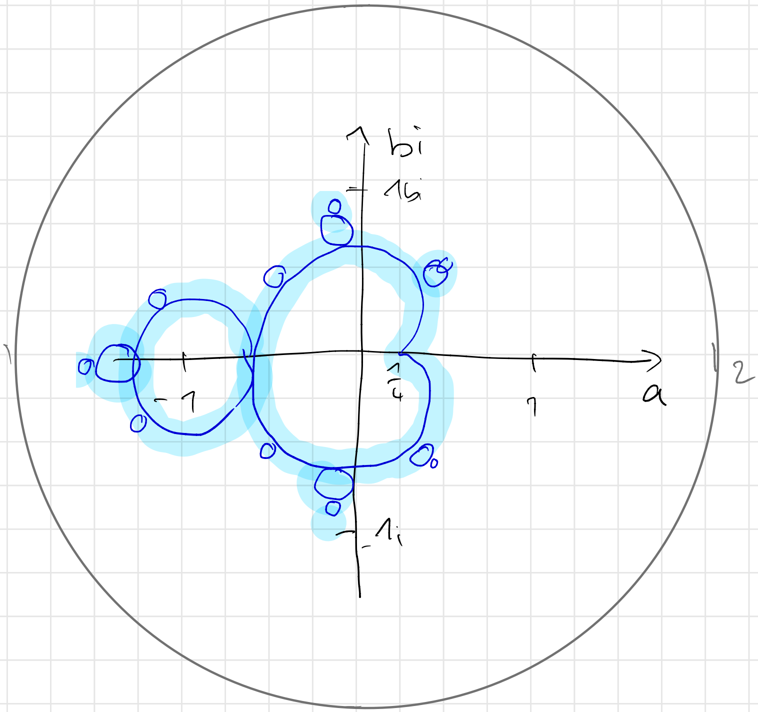
$f_{-1}(0) = 0^2 - 1 = -1$

⋮

CASE 2:
$|z|$ **bounded**
(≤ 2)

Check for each c in the plane.

The Mandelbrot set M is the set of complex numbers for which the distance $|z|$ (from the origin 0) is bounded.



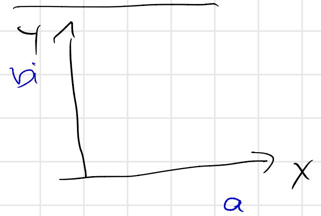
Coloring: „how quickly f escapes"

Boundaries:
- Infinitly detailled a every scale
- Self-similar but not quite repeating
- Formally incomputable in places

↳ if you chance c a little bit you can't predict what will happen on the boundary, while the insides are completly predictable.

„Square a number add a constand, repeat – but it encodes more visual and matematical complexity than most objects do."

Rendering



Pixel (p_x, p_y) corresponses to which complex number $(a+bi)$?

$$z = a + bi$$
$$c = x + yi$$

$$z^2 = (a+bi)^2$$
$$\Leftrightarrow (a+bi)(a+bi)$$
$$\Leftrightarrow a^2 + abi + abi - b^2$$
$$\Leftrightarrow (a^2 - b^2) + 2abi$$

$$\mid (bi)^2 = b^2 \cdot (-1)$$
$$= -b^2$$

$$z^2 + c = (a^2 - b^2) + 2abi + x + yi$$
$$\Leftrightarrow \underbrace{(a^2 - b^2 + x)}_{\text{real}} + \underbrace{(2ab + y)i}_{\text{imaginary}}$$

Pseudo Code

```
a = 0        x = ...  } from pixel coordinates
b = 0        y = ...

for i = 0 to max_iterations:
    a_next = a^2 - b^2 + x
    b-next = 2ab + y

    if (a_next^2 + b_next^2 > 4) BREAK
    a = a_next
    b = b-next
```